

Program Structure



General:

Live Teaching - Live Teaching sessions from Monday – Saturday from 08:45 AM to 05:45 PM.

For the Milestone(quiz and project), the submission timeline will be

Graded Quiz- 45 minutes

Graded Project- 2 hours

Course / Phase	Day No.	Date	Day	Module	Sub Topics
Core Java & Java Fundamentals (Phase 1)	Day - 1	15 April 2026	Wednesday	Git	Git intro, Create & Manage repositories, Repository settings, Adding & Committing, Branches, Stashing, GitHub, Fetching & Pulling, Git Collaboration, Git tags, Git Clone, Git Forking, Merge Requests
				Maven	Maven build and dependency management: POM structure, standard directory layout, build lifecycle & phases, dependency scopes, plugins & goals, packaging types (jar, war, ear), resource filtering, profiles, multi-module projects, repository management, versioning strategies, application packaging for servers and cloud runtimes, CI/CD integration
	Day - 2	16 April 2026	Thursday	Java Fundamentals	Java Architecture, Language Basics (JDK/JRE/JVM), Flow Control, Statements – ForEach, Arrays – Parallel Array Sorting, Java Features Primitive Data types, variables, Operators, String, StringBuilder, String API, Date/Time & numeric objects, Conditions, Switch, Loops & branches, Enum
	Day - 3 & 4	17 April 2026 & 18 April 2026	Friday & Saturday	OOPs / Inheritance / Abstraction	Classes and Objects, Encapsulation, Abstraction, Inheritance, Overriding, Polymorphism, Garbage Collection String, StringBuffer, Optional Class, StringJoiner Class Abstract Class, Interface – static and default methods, Final Keyword Introduction to Packages, Package imports Composition, Overloading and Overriding, Aggregation
	Day - 5	20 April 2026	Monday	Exception Handling & Packages	Introduction to Exception Handling, Exception Types, Try-Catch Blocks, Throws clause, Throw clause, Finally block, Custom exceptions, Catching Multiple Exceptions, Try-with-Resources, Error Hierarchy SOLID Design Principles
	Day - 6	21 April 2026	Tuesday	JUnit / Code Coverage	Introduction to JUnit 5, JUnit with Eclipse / VS Code, Assert methods and annotations, Test Suite, Parameterized Tests, Code Coverage tools
				Wrapper Classes	Wrapper Classes – boxing, unboxing, auto-boxing; Integer, Double, Character, Boolean classes and their utility methods
	Day - 7	22 April 2026	Wednesday	Collections / Stream API / Java 8 Features	Introduction to Collections Framework: List, Set, Map, Queue Interface, Deque Interface, Stack Class, Iterator Interface Sorting and Searching, Comparators and Comparables, Generics and Type Safety, Concurrent Collections, Synchronization and Thread Safety, Thread Local Java 8: Lambda Expressions, Stream API & Operators, Functional Interfaces, Default Methods in Interfaces, Method References, Optional Class, Date and Time API (java.time), Executor Service
	Day - 8	23 April 2026	Thursday	Design Patterns	Common Design Patterns: Singleton, Factory, Observer, Builder, Prototype, Adapter, Decorator, Strategy Applying Design Patterns in Java; Coding Best Practices, Code Review tools
	Day - 9	24 April 2026	Friday	Concurrency in Java	Multithreading Basics – Multi-Tasking OS understanding, Threads, Thread Scheduler, Thread Synchronization, Different ways of achieving multi-threading, Priority concept, Thread grouping, Intercommunication of Threads Java's Concurrency Utilities (java.util.concurrent), Thread Safety and Best Practices
	Day - 10	25 April 2026	Saturday	Java Latest Features (Java 9–23)	New Features in Java 9–23, deprecated APIs per version, key features in Java 11, 17, 21, 23 (Records, Sealed Classes, Pattern Matching, Text Blocks, Virtual Threads, Switch Expressions)
	Day - 11	27 April 2026	Monday	Milestone Assessment - 1 (Core JAVA)	
Data Base & Web Basics (Phase 2)	Day - 12 & 13	28 April 2026 & 29 April 2026	Tuesday & Wednesday	RDBMS/SQL/JDBC	Introduction to RDBMS, Oracle Introduction, Select Statement, Restricting and Sorting Data, DML, DDL, Introduction to JDBC, Establishing Connection, Executing Query and Processing result, Using PreparedStatement and Metadata objects, Using Callable Statement and Transactions
	Day - 14	30 April 2026	Thursday	HTML/CSS/Bootstrap	HTML Templates, Semantic HTML, multimedia, SVG, Geolocations, Web Storage, Accessibility (A11y), Inline, internal and external css, box model, style attribute, CSS Preprocessors, CSS Grid Layout, CSS-in-JS, Animation and Transitions, Responsive Design Patterns, Grid system, Typography, Navbars, button enhancements, Customization, SASS Integration, Advanced Components, Responsive Design, Accessibility in Bootstrap.
	Day - 15 & 16	2 May 2026 & 4 May 2026	Saturday & Monday	JavaScript	Date and Time Objects, Closures, DOM Objects, ES6+ Features, Promises and Async Programming, Closures and Scoping, Functional Programming, Error Handling, client side scripting, JSON, JSON - Objects, arrays etc, JSON-Ajax, Advanced DOM Manipulation, jQuery Plugins, Event Delegation, Ajax and Deferred Objects, Optimization and Performance
	Day - 17 & 18	5 May 2026 & 6 May 2026	Tuesday & Wednesday	Servlet/JSP	Introduction to Servlets, Introduction to JSP, Jsp Scripting Elements ,Implicit Objects , Jsp Actions , Javabeen in JSP
	Day - 19 & 20	7 May 2026 & 8 May 2026	Thursday & Friday	Spring Core	Introduction to Spring framework, Benefits of Spring, Features of Spring, Architecture, Bean Factory, Application Context, Spring Beans Scope, Dependency Injection/Inversion of Control, Setter based DI, Constructor based DI, Autowiring of Beans - By Type, By Name, By Constructor, Annotations for Autowiring
	Day - 21	09 May 2026	Saturday	Spring MVC	Introduction to 'Spring MVC' module, Front Controller, Components, Dispatcher Servlet, Handler Mapping bean, View Resolver bean, Understanding the flow of Spring MVC, Handler Mapping, Controller, Validators, MVC Form
	Day - 22 & 23	11 May 2026 & 12 May 2026	Monday & Tuesday	Object Relational Mapping	Define Persistence, Identify the problems with OR Mismatch, Describe alternatives for Object/Relational Mapping(ORM), Explain Hibernate as a preferred solution for ORM, Small demo on JDBC program for inserting a record, Program and verify a java application using hibernate framework

Java Web & Spring (Phase 3)				Hibernate Architecture	Explain Hibernate Architecture, Learn the Hibernate API- Configuration,SessionFactory,Session, Describe the details of OR Mapping, Explain the Hibernate Object States, Make Objects persistent, Retrieve a persistent object, Modify delete objects, CRUD operations
	Day - 24 & 25	13 May 2026 & 14 May 2026	Wednesday & Thursday	Querying in Hibernate	How to query with hibernate query language (HQL), How to query with Criteria API, Creating queries with Native SQL.
				Spring with Hibernate	Integrating Spring with Hibernate
	Day - 26 & 27	15 May 2026 & 16 May 2026	Friday & Saturday	Spring Boot	Spring boot and its benefits, Spring Boot Dependencies, Auto-Configuration & Runtime, Actuators, DevTools, Spring Boot starters thymeleaf views, Spring REST, Introduction, advantages, implementing REST and HTTP methods, Case study demo with all methods, Maven Dependency, Auto Generated credentials, Spring Security Filter, Spring Security Configuration, Ant, Matchers, Basic Auth Token, JWT, Exception Handling on REST end point.
	Day - 28	18 May 2026	Monday	Spring Data JPA	Overview, JDBC template API, Data Source, Configuration, RowMapper Interface, JdbcTemplate Example. Entity Mapping, Id Mapping, Columns Mapping, Inheritance Mapping, Embedded (Composition) Mapping, Association Mapping, JPQL. Embedded H2 and Embedded MongoDB, CRUD/JPA/MONGO Repository and custom implementation, Spring Test – unit testing (repository, API, Service), Spring Test Mocking, Introduction to API documentation Swagger
Microservice & Spring Cloud (Phase 4)	Day - 29 & 30	19 May 2026 & 20 May 2026	Tuesday & Wednesday	Introduction to Microservices	Kind of Applications, SOA based solutions, Microservices Architecture, Important attributes of Microservices, Thought behind Microservices, Monolith – features and challenges, Advantages and Disadvantages of Microservices, Microservices Frameworks, Best Practices, Design Patterns, Architecture, 12 Factor methodology and its implications on Microservices architecture
				Monolith Application	Creating a monolith application – Sample App, Redesigning monolith application into Microservices.
				Netflix OSS & Spring Cloud	What is Netflix OSS?, What is Spring Cloud?, Service Discovery with Spring Cloud Netflix Eureka, Client-Side Load Balancing with Spring Cloud Load Balancer, Implementing Microservices along with Eureka and Load Balancer, OpenFeign – declarative RestClient.
				Client-Side Circuit Breakers – Spring Cloud Circuit Breaker with Resilience4J	Understand how software circuit breakers protect against cascade failure, Use Spring Cloud Circuit breaker with Resilience4J to implement circuit breakers, Establish simple monitoring of circuit breakers using Prometheus dashboard.
				Other Spring Cloud Projects	Spring Cloud Configuration Server and Client, API Gateway - Spring Cloud Gateway, Distributed Log tracing with Spring Cloud Sleuth and Zipkin, Introduction to Spring Security, Basic Auth
				Event Driven Microservices with Apache Kafka	Introduction to Kafka, Kafka architecture, Apache Kafka server installation, Kafka message producer and consumer, Topic and partition, Consuming messages from Spring Boot/Microservices application, Communication between Microservices using Kafka server
React JS (Phase 4)	Day - 31 & 32	21 May 2026 & 22 May 2026	Thursday & Friday	Introduction to React JS	Virtual DOM concepts, JSX concepts.
				create-react-app	React Project generation and Build Tool, Structure of the React Application.
				React Component	Class and Functional components, Event Handling.
				React State	Controlled Components.
	Day - 33 & 34	23 May 2026 & 25 May 2026	Saturday & Monday	Props(properties) Using refs	Passing arguments between the components, Default property initialization, PropTypes, Uncontrolled Components.
				Life Cycle	Component Life Cycle methods.
				Composable Components	Multiple components interacting each other, Reading the data from in-memory, React application interfacing with backend server for data (json-server).
				Form handling	Form handling using Formik, Yup.
				Higher Order Components, Render Props	Understanding HOC Concepts, Implementing simple HOC Component.
				React Context API & React Router	Context API & React Router concepts.
				Routing Transition	Routing Transition Concepts (Transition into and Transition from).
	Day - 35	26 May 2026	Tuesday	Milestone Assessment - 2 (React)	
GenAI & Copilot (Phase 4)	Day - 36	28 May 2026	Thursday	Generative AI	LLM fundamentals, Prompt engineering, Embeddings and vector databases, RAG architecture, AI agents
	Day - 37	29 May 2026	Friday	GitHub Copilot	AI security and privacy, GenAI integration patterns in Java applications, Enterprise GenAI architectures
					Developer-oriented prompt patterns, GenAI testing and validation basics, Responsible AI usage
	Day - 38	30 May 2026	Saturday	Final Milestone Assessment (MCQ all topics including Gen AI + coding on Core topic)	
CAPSTONE	Day - 39– 43	1 June 2026 - 5 June 2026	Monday - Friday	Full-Stack Capstone Project	